**bi:** Trong khũng gian còc ma trn vuũng cp hai, xỗt xem h 3 vect $A, B, C$ sau óy c lp tuyn tònh hay ph thuc tuyn tònh:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 6 \\ 1 & 6 \end{bmatrix}$$

Xỗt phng trønh t hp tuyn tònh ca h vect bng vect khũng :

$$x_1 A + x_2 B + x_3 C = \mathbf{0}$$

Thay còc ma trn vịo, ta c:

$$x_1 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} + x_2 \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix} + x_3 \begin{bmatrix} 2 & 6 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Cng còc ma trn v trời vị ng nht còc phn t tng ng, ta thu c h phng trønh thun nht:

$$\begin{cases} x_1 + 2x_3 = 0 \quad (1) \\ 2x_1 + 2x_2 + 6x_3 = 0 \quad (2) \\ x_2 + x_3 = 0 \quad (3) \\ x_1 + 4x_2 + 6x_3 = 0 \quad (4) \end{cases}$$

Gii h phng trønh:

T $(1) \Rightarrow x_1 = -2x_3$

T $(3) \Rightarrow x_2 = -x_3$

Thay $x_1$ vị $x_2$ vịo phng trønh $(2)$: $2(-2x_3) + 2(-x_3) + 6x_3 = -4x_3 - 2x_3 + 6x_3 = 0$ (Luũn ũng).

Thay $x_1$ vị $x_2$ vịo phng trønh $(4)$: $(-2x_3) + 4(-x_3) + 6x_3 = -6x_3 + 6x_3 = 0$ (Luũn ũng).

H phng trønh cú vũ s nghim ph thuc vịo $x_3$.

Vơ h ph thuc vịo $x_3$ nổn ch cn chn $x_3$ bt kơ thơ ta luũn cú 1 nghim khũng tm thng $\Rightarrow$ Tn ti còc h s khũng ng thi bng 0 sao cho t hp tuyn tònh bng $\mathbf{0}$, nổn h vect $\{A, B, C\}$ lị ph thuc tuyn tònh.

**Gii Bi 3**

**bi:** Cho tp hp $S = \{(\alpha - \beta, 2\alpha + \beta, \alpha, \beta) \mid \alpha, \beta \in \mathbb{R}\}$. Chng minh $S$ lị khũng gian con ca $\mathbb{R}^4$. Tơm c s vị s chiu ca $S$.

**1. Chng minh $S$ lị khũng gian con:**

$\forall u \in S$ u cú dng $u = (\alpha - \beta, 2\alpha + \beta, \alpha, \beta)$. Ta cú th phón tòch:

$$u = (\alpha, 2\alpha, \alpha, 0) + (-\beta, \beta, 0, \beta)$$

$$\Rightarrow u = \alpha(1, 2, 1, 0) + \beta(-1, 1, 0, 1)$$

t $v_1 = (1,2,1,0)$ vị $v_2 = (-1,1,0,1)$. Khi ú, mi vect ca $S$ u lị t hp tuyn tờnh ca $v_1$ vị $v_2$.
Núi còch khòc, $S = \text{span}\{v_1, v_2\}$.
Vơ $S$ lị khũng gian sinh bi mt tp hp còc vect trong $\mathbb{R}^4$, nổn theo nh ngha, $S$ lị mt khũng gian con ca $\mathbb{R}^4$.

**2. Tơm c s vị s chiu ca $S$:**
H sinh ca $S$ lị $F = \{v_1, v_2\}$. Xỗt phng trơnh:

$$k_1 v_1 + k_2 v_2 = (0,0,0,0)$$

$$\Rightarrow (k_1 - k_2, \quad 2k_1 + k_2, \quad k_1, \quad k_2) = (0,0,0,0)$$

$\Rightarrow k_1 = 0$ vị $k_2 = 0$.
$\Rightarrow$ h $F = \{v_1, v_2\}$ c lp tuyn tờnh.
Mt c s ca khũng gian con $S$ lị $B = \{(1,2,1,0), (-1,1,0,1)\}$. S chiu $\dim(S) = 2$.

**Gii Bịi 11**
**bịi:** Khũng gian $P_3$ gm a thc bc $\leq 3$. Xỗt hai h vect $(\alpha)$ vị $(\beta)$.

**a) Chng minh h $(\alpha)$ vị $(\beta)$ lị hai c s ca $P_3$:**
Khũng gian $P_3$ cú s chiu $\dim(P_3) = 4$.
H $(\alpha) = \{1, x, x^2, x^3\}$ gm 4 a thc c lp tuyn tờnh (c s chờnh tc ca $P_3$), do ú $(\alpha)$ lị mt c s.
chng minh $(\beta)$ lị c s, ta biu din còc a thc ca $(\beta)$ qua c s $(\alpha)$:

$$\beta_1 = 1 = \mathbf{1} \cdot 1$$

$$\beta_2 = x - 2 = -\mathbf{2} \cdot 1 + \mathbf{1} \cdot x$$

$$\beta_3 = (x-2)^2 = x^2 - 4x + 4 = \mathbf{4} \cdot 1 - \mathbf{4} \cdot x + \mathbf{1} \cdot x^2$$

$$\beta_4 = (x-2)^3 = x^3 - 6x^2 + 12x - 8 = -\mathbf{8} \cdot 1 + \mathbf{12} \cdot x - \mathbf{6} \cdot x^2 + \mathbf{1} \cdot x^3$$

Ma trn ta theo ct ca h $(\beta)$ i vi c s $(\alpha)$ lị:

$$P = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 0 & 1 & -4 & 12 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Cú nh thc $\det(P) = 1 \cdot 1 \cdot 1 \cdot 1 = 1 \neq 0$. Suy ra h $(\beta)$ c lp tuyn tờnh. Mị h cú ững 4 phn t bng s chiu ca khũng gian, nổn $(\beta)$ cng lị mt c s ca $P_3$.

**b) Tơm ma trn chuyn t c s $(\alpha)$ sang c s $(\beta)$:**

Theo kt qu tònh toòn cóu a:

$$P_{\alpha\to\beta} = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 0 & 1 & -4 & 12 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

**c) Tơm ta ca vect $v = x^3 - 3x + 2$ i vi c s $(\beta)$:**

Gi s ta ca $v$ i vi $(\beta)$ lị $[v]_\beta = (c_1, c_2, c_3, c_4)^T$. Khi ú:

$$x^3 - 3x + 2 = c_1(1) + c_2(x-2) + c_3(x-2)^2 + c_4(x-2)^3$$

Khai trin v phi theo ly tha ca $x$ vị nhúm li:

$$x^3 - 3x + 2 = c_4x^3 + (c_3 - 6c_4)x^2 + (c_2 - 4c_3 + 12c_4)x + (c_1 - 2c_2 + 4c_3 - 8c_4)$$

ng nht h s hai v, ta c h phng trơnh:

$$\begin{cases} c_4 = 1 \\ c_3 - 6c_4 = 0 \\ c_2 - 4c_3 + 12c_4 = -3 \\ c_1 - 2c_2 + 4c_3 - 8c_4 = 2 \end{cases}$$

Gii t trổn xung di:

$c_4 = 1$

$c_3 = 6(1) = 6$

$c_2 = -3 + 4(6) - 12(1) = 9$

$c_1 = 2 + 2(9) - 4(6) + 8(1) = 4$

Vy ta cn tơm lị:

$$[v]_\beta = \begin{bmatrix} 4 \\ 9 \\ 6 \\ 1 \end{bmatrix}$$

**Gii Bị 5**

**a) Chng minh $f$ lị ònh x tuyn tònh**

$\forall x, y \in \mathbb{R}^4 : x = (x_1, x_2, x_3, x_4); y = (y_1, y_2, y_3, y_4); \forall k \in \mathbb{R}$

$x+y=(x_1+y_1,x_2+y_2,x_3+y_3,x_4+y_4)$, ta cú:

$$\begin{aligned}
f(x+y) &= f(x_1+y_1,x_2+y_2,x_3+y_3,x_4+y_4)\\
&= (2(x_1+y_1)+2(x_2+y_2)-(x_3+y_3),(x_2+y_2)+(x_3+y_3)+(x_4+y_4),(x_3+y_3)+2(x_4+y_4))\\
&= ((2x_1+2x_2-x_3)+(2y_1+2y_2-y_3),(x_2+x_3+x_4)+(y_2+y_3+y_4),(x_3+2x_4)+(y_3+2y_4))\\
&= (2x_1+2x_2-x_3,x_2+x_3+x_4,x_3+2x_4)+(2y_1+2y_2-y_3,y_2+y_3+y_4,y_3+2y_4)\\
&= f(x)+f(y)
\end{aligned} \tag{1}$$

$$\begin{aligned}
f(kx) &= f(kx_1,kx_2,kx_3,kx_4)\\
&= (2(kx_1)+2(kx_2)-(kx_3),(kx_2)+(kx_3)+(kx_4),(kx_3)+2(kx_4))\\
&= (k(2x_1+2x_2-x_3),k(x_2+x_3+x_4),k(x_3+2x_4))\\
&= k(2x_1+2x_2-x_3,x_2+x_3+x_4,x_3+2x_4)\\
&= kf(x)
\end{aligned} \tag{2}$$

T (1) vị (2) ta suy ra $f$ lị ònh x tuyn tờnh.

**b) Tơm ma trn ca $f$ trong cp c s chờnh tc.**

Ta gi $\{e_1=(1,0,0,0);e_2=(0,1,0,0);e_3=(0,0,1,0);e_4=(0,0,0,1)\}$ lị c s chờnh tc trong $\mathbb{R}^4$.
$\{e'_1=(1,0,0);e'_2=(0,1,0);e'_3=(0,0,1)\}$ lị c s chờnh tc trong $\mathbb{R}^3$, nh vy:

$$\begin{aligned}
f(e_1) &= f(1,0,0,0)=(2,0,0)=2e'_1+0e'_2+0e'_3\\
f(e_2) &= f(0,1,0,0)=(2,1,0)=2e'_1+1e'_2+0e'_3\\
f(e_3) &= f(0,0,1,0)=(-1,1,1)=-1e'_1+1e'_2+1e'_3\\
f(e_4) &= f(0,0,0,1)=(0,1,2)=0e'_1+1e'_2+2e'_3
\end{aligned}$$

Suy ra ma trn $A$ (vi còc ct lị ta  ca nh còc vect c s) lị:

$$A=\begin{bmatrix}2&2&-1&0\\0&1&1&1\\0&0&1&2\end{bmatrix}$$

**c) Tơm** $\ker(f),\mathbf{Im}(f)$**, c s vị s chiu ca** $\ker(f),\mathbf{Im}(f)$**.**

**Tơm** $\ker(f)$**:**

Theo nh ngha, $x\in\ker(f)\Leftrightarrow f(x)=(0,0,0)$.

Ta cú h phng trơnh thun nht:

$$\begin{cases} 2x_1 + 2x_2 - x_3 = 0 & (1) \\ x_2 + x_3 + x_4 = 0 & (2) \\ x_3 + 2x_4 = 0 & (3) \end{cases}$$

Gii h phng trơnh trổn, t phng trơnh (3) suy ra $x_3 = -2x_4$.

Thay vịo phng trơnh (2): $x_2 - 2x_4 + x_4 = 0 \Rightarrow x_2 = x_4$.

Thay vịo phng trơnh (1): $2x_1 + 2(x_4) - (-2x_4) = 0 \Rightarrow 2x_1 + 4x_4 = 0 \Rightarrow x_1 = -2x_4$.

Nghim ca h lị $x = (-2x_4, x_4, -2x_4, x_4) = x_4(-2, 1, -2, 1)$ vi $\forall x_4 \in \mathbb{R}$.

Vy $\ker(f) = \{(-2b, b, -2b, b) \mid b \in \mathbb{R}\}$.

C s ca $\ker(f)$ lị $\{(-2, 1, -2, 1)\}$ vị s chiu $\dim(\ker(f)) = 1$.

**Tơm Im$(f)$:**

Khũng gian nh $\mathrm{Im}(f)$ c sinh bi còc ct ca ma trn $A$.

$$\mathrm{Im}(f) = \mathrm{Span}\{(2,0,0), (2,1,0), (-1,1,1), (0,1,2)\}$$

Ma trn $A$ õ dng bc thang vi 3 phn t c s khòc khũng (ti ng chỗo chờnh), do ú hng ca ma trn bng 3.

T nh lỳ v s chiu:

$$\dim(\mathrm{Im}(f)) = \dim(\mathbb{R}^4) - \dim(\ker(f)) = 4 - 1 = 3$$

Vơ $\dim(\mathrm{Im}(f)) = 3$ bng vi s chiu ca khũng gian ờch $\mathbb{R}^3$, nổn tp 3 vect ct bt k c lp tuyn tờnh s to thnh mt c s. Chng hn, mt c s ca $\mathrm{Im}(f)$ lị $\{(2,0,0), (2,1,0), (-1,1,1)\}$.